能量部分：

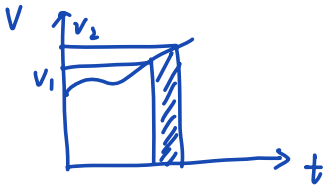
$\Delta t \to 0$

$(v_2+v_1)(v_2-v_1)/2a$

$= \frac{(v_2+v_1)}{2}\Delta t = \int_{t_1}^{t_2} v\,dt$

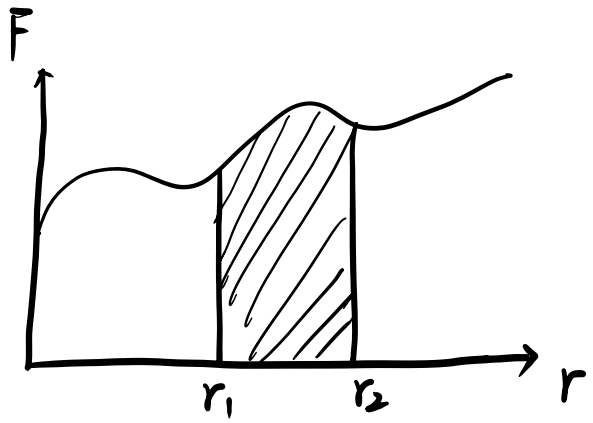
$$v_2^2 - v_1^2 = 2a(r_2 - r_1)$$

$$\frac{1}{2} m(v_2^2 - v_1^2) = ma(r_2 - r_1)$$



$$E_{k2} - E_{k1} = F\Delta r$$

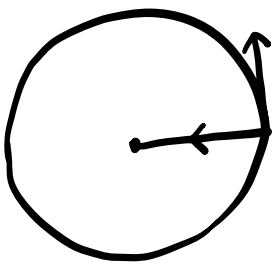
动能的变量 ，外界做功。



$\vec{F}\cdot d\vec{r}$ 做功微元

$\vec{F}(\vec{r})$ 力在力的方向上位移的积累

匀速圆周运动，功能原理



$$\Delta W = \int \vec{F}\cdot d\vec{r}$$

$$= \Delta E_k = 0$$

时时垂直

$$\int F\,dx = \int -kx\,dx = -\frac{1}{2}kx^2\Big|_{x_1}^{x_2}$$



$$= \frac{1}{2}k(x_1^2 - x_2^2)$$

$$E_{k2} - E_{k1} = \frac{1}{2}k(x_1^2 - x_2^2) \qquad \Delta E_p = -\Delta E_k$$

$$\frac{1}{2}kx_1^2 + E_{k1} = \frac{1}{2}kx_2^2 + E_{k2} \qquad = -F\Delta x$$

机械能守恒.

非保守力

friction: $\int F_f\,dx$ (速度反向, F_f 反向)

F_f 不是位置 x 的函数

(电磁相互作用)

$F = -kx$ (position dependent)

Conservative force $\Rightarrow$

conservation

和与路径无关 (两点确定三维)

potential :

$$\int_A^B dE_p = \ominus \int_A^B F(x)\,dx$$

↓

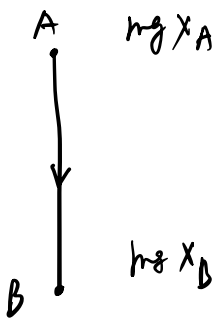
$dE_p = -F\,dx$　　势能的增加对应势能减少

$$\Delta E_p = -F(x)\,\Delta x \qquad \vec{F}(x) = m\vec{g} \downarrow$$

$$E_{pB} - E_{pA} = -\int_A^B F(x)\,dx$$

自带符号

$$= +mg(x_B - x_A)$$



↑ (竖直向上为正方向)

$$E_{pB} = mgx_B + C$$

$$E_{pA} = mgx_A + C$$

$$E_p(y) = -\int_0^y F\,dx$$

$$= mgy$$

约定　$x=0$ 处 势能为 0

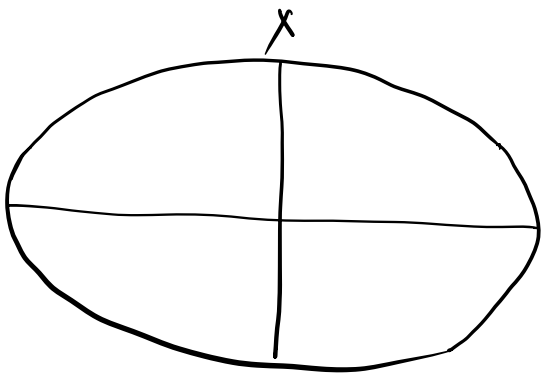
$$-\int_0^X F\,dx = \frac{1}{2}kX^2$$

变化量为 X，无论伸长或缩短

弹的势能增为 $\frac{1}{2}kX^2$

$$E = \frac{1}{2}mv^2 + \frac{1}{2}kX^2 \Rightarrow$$

$$\frac{v^2}{2E/m} + \frac{X^2}{2E/k} = 1$$



弹簧振动

消掉 t

参数方程

椭圆上每一点对应相同能量

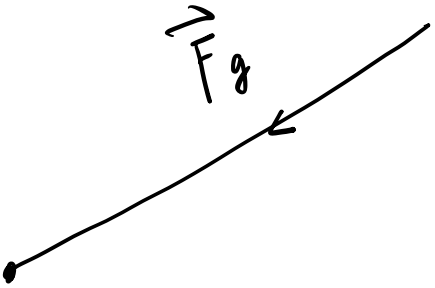
more general : 做功 $\longrightarrow$ 动能改变

引力做功： 方向导数，引力势能

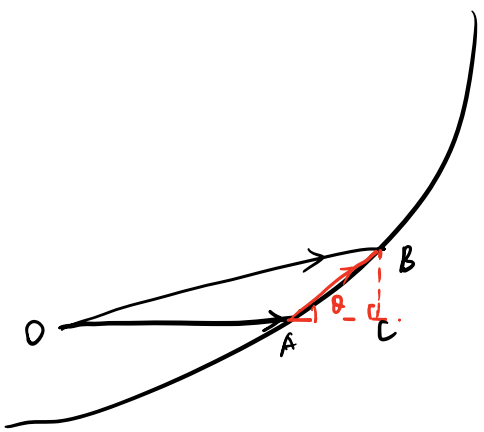
$$\vec{F}_g = -\frac{GMm}{r^2}\vec{e}_r \quad (\text{吸引力})$$

问题：

（引力势能定义在哪？）



$$\vec{F}_g = -\nabla E_p = -\frac{GMm}{r^2}\vec{e}_r$$



几何

$$\int dw = \int_A^B \vec{F}(r)\cdot d\vec{r}$$

引力做功微元

$$= \int_A^B - \frac{GMm}{r^2} \boxed{d\vec{r}\cos\theta}$$

(吸引力)　　$= dr$

$B \to A$

$dr = |OB| - |OA| \approx$ in the limit, $OC = OB$

$|OC| - |OA| = |AC|$　　$AC = |d\vec{r}|\cos\theta = dr$

$$= \int_A^B - \frac{GMm}{r^2} dr$$

$$= \int_A^B d\left(\frac{GMm}{r}\right)$$

$$E_p\Big|_A^B = -\int_A^B d\left(\frac{GMm}{r}\right)$$

代数处理

$$E_p^B - E_p^A = -\int_A^B \vec{F} \cdot d\vec{r}$$

$$= -\int_A^B (-)\frac{GMm}{r^2}\,\vec{e}_r \cdot d\vec{r}$$

$$= +\int_A^B \frac{GMm}{r^3}\left[r\,\vec{e}_r \cdot d\vec{r}\right]$$

$$\left[(x,y,z)\cdot(dx,dy,dz)\right]$$

$$x\,dx + y\,dy + z\,dz$$

$$\frac{1}{2}\,d\left(x^2+y^2+z^2\right) = \frac{1}{2}\,dr^2$$

$$= r\,dr$$

$$= +\int_A^B \frac{GMm}{r^2}\,dr$$

$$= \int_A^B d\left(-\frac{GMm}{r}\right)$$

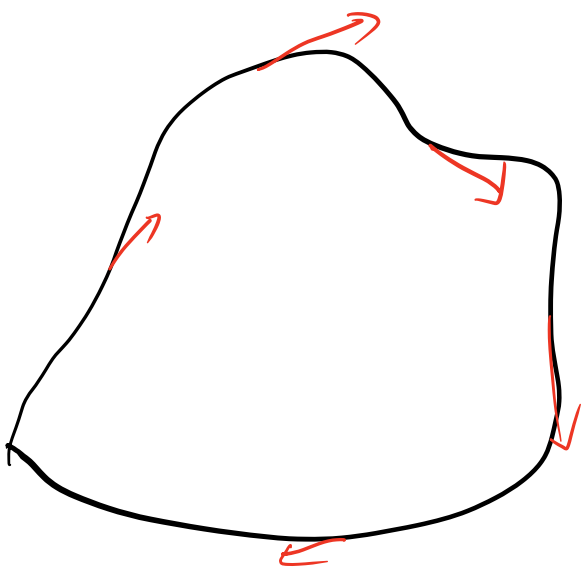
$$E_p = -\frac{GMm}{r}$$

$$\int_A^B (F_c + \vec{F}_{nc})\,dx = \Delta W = \Delta E_k$$

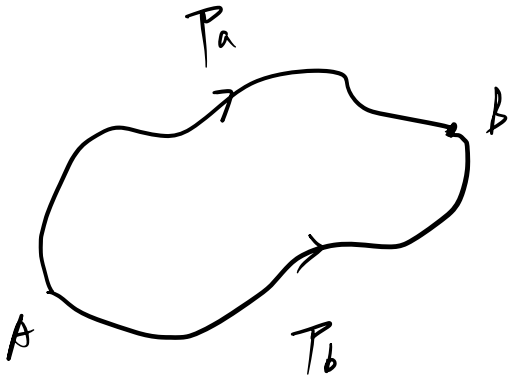
非保守力

$$W_{nc} + U(A) - U(B) = E_k(B) - E_k(A)$$

$$W_{nc} = \Delta(E_k + U)$$



$$\oint \vec{F}_c \cdot d\vec{r} = 0$$



$$\int_A \vec{F} \cdot d\vec{r} = \int_B \vec{F} \cdot d\vec{r}$$

$$\int_A \vec{F} \cdot d\vec{r} - \int_B \vec{F} \cdot d\vec{r}$$

$$= \oint \vec{F} \cdot d\vec{r} = 0$$

Relation between conservative force and potential energy

In 1D: $U(x_2) - U(x_1) = -\int_1^2 F_c \, dx$

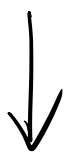
$$F_c = -\frac{dU}{dx}$$

$$\text{例}: \quad ① \quad U_G = mgz + \text{const}$$

$$F_G = -\frac{dU_G}{dz} = -mg$$

$$② \quad U_S = \frac{1}{2}kx^2 + \text{const}$$

$$F_S = -\frac{dU_S}{dx} = -kx$$



generalize to 3D

$$\vec{F}_C = \left(-\frac{\partial U}{\partial x}, -\frac{\partial U}{\partial y}, -\frac{\partial U}{\partial z}\right)$$

$$F_x = -\frac{\partial U}{\partial x} = GMm\frac{\partial}{\partial x}\left(\frac{1}{r}\right)$$

$$= GMm\left(-\frac{1}{r^2}\right)\frac{\partial\left(x^2+y^2+z^2\right)^{\frac{1}{2}}}{\partial x}$$

$$= \sim \frac{\frac{1}{2}(2x)}{r}$$

$$F_y = \sim \frac{y}{r}$$

$$F_z = \sim \frac{z}{r}$$

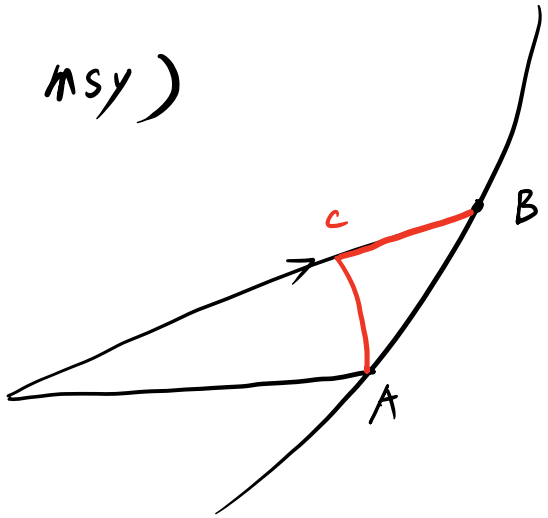
$$F = -\frac{GMm}{r^3}\underbrace{(x, y, z)}_{\text{坐标}}$$

$$= -\frac{GMm}{r^2}\vec{e}_r$$

坐标前的矢量确定方向

学生新思路

（张嘉 2020 MSY）



$$\Delta W = \int_A^B \vec{F} \cdot d\vec{r}$$

$$= \int_{ACB} \vec{F} \cdot d\vec{r}$$

$$= \int_A^C \vec{F} \cdot d\vec{r} + \int_C^B \vec{F} \cdot d\vec{r}$$

$\vec{F} \perp d\vec{r}$，$W = 0$

$\int_C^B \vec{F} \cdot d\vec{r}$ 恒向

Note：很多重点书在这里，

$$W = \int_A^B \vec{F} \cdot d\vec{r}$$

$\vec{F} = m\vec{g} = -mg$ （竖直向下）

$A \to B$ 下降

如果是自由落体，$d\vec{r}$ 在一维路径下

写成 dr 是自然符号

依赖于所选路径，dr 有正有负

同理 $F = -kx$

x 是位移，x 本身携带符号

我们处理引力做功问题：

① 几何

② 代数

③ 路径选择

Feynmann lectures:

the forces between the particles at the most fundamental level, are conservative

When friction is present it is not true that kinetic energy is lost, even though a sliding object stops and the kinetic energy seems to be lost. The kinetic energy is not lost because, of course, the atoms inside are jiggling with a great amount of kinetic energy than before, and although we cannot see that, we can measure it by determining the temperature.